

Muhammad Shurahbeel Bin Sajid

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RESEARCH INTERESTS

My research is in video understanding and structured representation learning. My thesis fused appearance and motion streams through cross-attention for human–object interaction recognition; separately, I worked on Traditional Chinese Medicine prescriptions as hypergraphs to capture multi-way combinations pairwise graphs cannot express. Both concern how interaction structure should be represented so models learn relations rather than co-occurrence statistics.

EDUCATION

M.S. Data Science

Sep 2023 – Nov 2025

Air University, Islamabad, Pakistan

- Thesis: Unifying Appearance and Motion: Transformer-Based Feature Fusion for HOI Recognition. Supervisor: Prof. Danish Hamid.
- Relevant coursework: Deep Learning, Machine Learning, Computer Vision, Big Data Analytics.
- *Visiting exchange student*, Northwestern Polytechnical University, Xi'an, China (Aug 2024 – Jan 2025), coursework and lab research on full scholarship.

B.S. Mathematics

Feb 2016 – May 2020

International Islamic University, Islamabad, Pakistan

- Relevant coursework: Real Analysis, Linear Algebra, Probability and Statistics, Numerical Methods, Calculus, Differential Equations.

PUBLICATIONS

TransFlow-HOI: Transformer-Driven Multimodal Fusion for High-Performance HOI Recognition

M. Shurahbeel B S, Danish H, Tahir A; et al., "TransFlow-HOI: Transformer-Driven Multimodal Fusion for High-Performance Human–Object Interaction Recognition," in *IEEE Access*, vol. 14, pp. 12717-12732, 2026, doi: 10.1109/ACCESS.2026.3655585.

A transformer-based framework fusing RGB and optical-flow streams through cross-attention to capture long-range temporal dependencies. Attains 93.1% top-1 accuracy on HOI4D, a 14.9-point improvement over Late Fusion 78.2% and 15.8 points over the Shallow MLP Fusion 77.3% baseline.

RESEARCH EXPERIENCE

Graduate Research Assistant

Nov 2023 – Apr 2025

Air University, Islamabad, Pakistan

- Designed **TransFlow-HOI**, a transformer architecture that fuses RGB appearance and optical-flow motion streams via cross-attention to model long-range temporal dependencies in human–object interaction video.
- Built the full experimental pipeline on the **HOI4D** benchmark: I3D feature extraction, optical-flow computation, two-stage and single-stage training regimes, and evaluation across accuracy, precision, recall, and F1.
- Surveyed the video understanding, action recognition, and object detection literature to position the contribution against I3D (Two-Stream Baseline), MS-TCN(Temporal Modeling), and ASFormer (Transformer-based Temporal Model).

Visiting Research Student, Bioinformatics R&D Lab

Sep 2024 – Jan 2025

Northwestern Polytechnical University, Xi'an, China

- Modelled Traditional Chinese Medicine prescription records as herb–symptom networks and as **hypergraphs**, capturing multi-herb combinations that pairwise graphs cannot represent.
- Cleaned and normalized heterogeneous herbal datasets for treatment prediction, resolving inconsistent nomenclature and dosage encoding across different records.
- Applied network-based pattern discovery to surface recurring co-prescription motifs and their symptom associations.

HONOURS AND AWARDS

- **Outstanding Visiting Student Scholarship**, Northwestern Polytechnical University, 2024; sole recipient nominated from Air University.

PROFESSIONAL EXPERIENCE

Jr. AI Engineer

Jul 2025 – Present

CareCloud (MTBC), Islamabad, Pakistan

- Build and deploy LLM/NLP pipelines that parse and summarize clinical documents, extract structured clinical fields, and feed downstream billing operations for U.S. healthcare revenue cycle management.
- Develop machine learning models for Prior Authorization and Denial Management workflows, covering feature engineering, evaluation design, and integration into production systems.
- Work with clinical and RCM domain experts to define evaluation criteria and validate model outputs against expert judgment.

AI Intern

May 2025 – Jul 2025

CareCloud (MTBC), Islamabad, Pakistan

- Fine-tuned open-source LLMs to build a medical document summarizer, reducing clinical review time by **40%** on internal evaluation sets.
- Contributed to the architecture of AI-driven automation modules subsequently taken to production deployment.

SELECTED PROJECTS

Crisis Weave — Multimodal Retrieval for Disaster Reporting

github.com/ShurahbeelSajid/Crisis-Weave

- Built an evidence-first retrieval-augmented QA system over heterogeneous disaster reports: native text, OCR output, document images, video frames and transcripts, and structured tables.
- Implemented citation validation, source attribution, and conflict and freshness detection, with explicit abstention when retrieved evidence is insufficient prioritizing calibrated refusal over confident hallucination.
- Integrated guarded read-only SQL and approved-domain web search for structured and external evidence, under tenant-isolated libraries.

AI Detection Suite — Real-Time Local Inference

github.com/ShurahbeelSajid/ai-detection-suite

- Developed a local-first webcam pipeline combining YOLO object detection with CPU-based Haar face detection and lightweight hand detection.
- Engineered bounded background inference, coordinate scaling, and deterministic resource cleanup for stable sustained throughput; instrumented an FPS and latency HUD for profiling.

TECHNICAL SKILLS

Languages

Python, MATLAB, SQL

Frameworks	PyTorch, TensorFlow, Keras, scikit-learn
Computer Vision	I3D feature extraction, optical flow, YOLO, OpenCV, image segmentation and analysis
Network Analysis	NetworkX, HyperNetX, Cytoscape, Gephi
Data	NumPy, pandas, pyspark, data collection, ETL & ELT, preprocessing, visualisation

ACADEMIC SERVICE AND LEADERSHIP

- **Host and Organizer**, International Academic Conference on AI and Machine Learning, Northwestern Polytechnical University, 2025
- **CEP International Talent Program**, School of Software, Northwestern Polytechnical University, collaborative R&D with international and Chinese researchers.
- Organized seminars and formal academic talks at Air University; represented the university in intervarsity debate and public speaking.

LANGUAGES

Urdu (native) | English (C1) | Turkish (A2) | Chinese (A1)

REFEREES

Prof. Danish Hamid

Assistant Professor, Air University
MS thesis supervisor
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Prof. Chen Bolin

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